DPUEnvSCNX

DPUEnvSCNX can be used to set environmental and lighting conditions in the simulation.

DPUEnvSCNX includes all functionality of **DPUEnvControl**. The shared aspects are described in its manual; please consult it additionally.

Extending the functionality of DPUEnvControl, DPUEnvSCNX additionally allows to modify the simulated environment via road events (*hedgehogs*) in the database SCNX.

All of these hedgehogs are defined within the family Env.

For each Variable of DPUEnvControl, the following hedgehogs are supported:

[VarName]	Changes the target value of the variable.
CR[VarName]	Modifies the variable's change rate (in 1/s).

A list of all Variables and Modules can be found in DPUEnvControl's manual.

1. Inputs:

In[VarName]	Input used to set the target value of variable *[VarName]*
ChangeRate[VarName]	Input to set the change rate of variable *[VarName]*

2. Dynamic parameters:

ShowDialog	Shows a dialog window (on the computer where DPUEn-vControl is running) which can be used to change the values of all Variables.

3. Dynamic parameters for each Module:

Auto[ModuleName]	Boolean value (0 or 1), which enables the automatic calculation of values of module [*ModuleName*].

4. Outputs:

Out[VarName]	Output which shows the current value of variable *[Var-Name]*

5. Dependencies to other DPUs:

The database **DPUSCNX** must run on each computer on which **DPUEnvSCNX** is used. The DPU index assigned to **DPUEnvSCNX** should be larger than the one assigned to **DPUSCNX**.

Either DPUEnvControl or DPUEnvSCNX should be run on each computer which simulates traffic, pedestrians, or provides a graphical view.

6. Example:

All hedgehog examples assume a module *Track* which contains a course instance *section*.

The basic definition is as follows:

```
define Module Track
{
	...
	#
	Hedgehogs =
	{	# (Course, course length, direction, lane ID, family, name,
	value)
		(section, 100, Normal, 0, Env, TimeOfDay, 0.0)
	};
	...
};
```

More than one Variable can be set at once. To do so, additional names and values are appended to the hedgehog. This is especially useful to modify target value and change rate of a Variable at once.

```
define Module Track
{
	...
	#
	Hedgehogs =
	{	# (Course, course length, direction, lane ID, family, name,
	value, ...)
		(section, 200, Normal, 0, Env, TimeOfDay, 16.0, CRTimeOfDay,
	0.5),
		(section, 500, Normal, 0, Env, TimeOfDay, 20.0, Humidity, 0.3)
	};
...
};
```